

Topos-Theoretic Bridges: Transferring Results Across Case-Theoretic Frameworks

Topos-Theoretic Bridges: Transferring Results Across Case-Theoretic Frameworks I

Where we are in the argument. ??–?? have erected four formal layers — typological categories (??), type-logical pregroup syntax (??–??), distributional semantics (??–??), and enriched magnitude (??–??). This chapter supplies the meta-framework that lets results proven in one layer transfer to the others: classifying toposes and Morita equivalence (with the honest limitation that the repository checks necessary-but-not-sufficient invariant matching, not full topos equivalence).

The Inter-Theoretic Translation Problem I

The preceding sections constructed four formally distinct perspectives on case: typological, type-logical, distributional, and enriched. A central methodological question arises: *when can structural results proved in one framework be carried over to another without starting from scratch?* Caramello's [-@caramello2016bridges] topos-theoretic bridge technique provides this methodology for properties that are **invariants of a shared classifying topos**, once Morita equivalence (or a suitable bridge topos) is in hand. We adopt this methodology as a research programme for case theory: establishing full Morita equivalences for the specific case-theoretic formalizations developed here remains largely open (see the implementation note later in this section), but the framework precisely identifies *what* must be proved and *what* would transfer as a result.

Classifying Toposes: The Logical Shape of a Theory I

A Topos Is a Self-Contained Logical Universe

A **topos** is a category possessing the structural richness of a generalized “universe of sets”: products, exponentials, and a subobject classifier (a “truth-value object”) that supports internal first-order reasoning. The most familiar example is the category **Set** of ordinary sets; other important instances include presheaf categories $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ and sheaf categories over topological spaces. Intuitively, a topos provides a *self-contained logical universe* within which mathematical reasoning can proceed—and different toposes encode different logical constraints.

Classifying Toposes: The Logical Shape of a Theory II

Morita Equivalence: Invariant Transfer Across Typologies

Every *geometric theory* $\mathbb{T}$ (axiomatized by sequents with finite conjunctions, arbitrary disjunctions, and existential quantification) generates a unique **classifying topos** $\mathcal{E}_{\mathbb{T}}$: a canonical topos such that, for any Grothendieck topos $\mathcal{F}$, models of $\mathbb{T}$ in $\mathcal{F}$ correspond naturally to **geometric morphisms** $\mathcal{F} \rightarrow \mathcal{E}_{\mathbb{T}}$. The classifying topos encodes the theory's “logical shape” independently of any particular model.

Caramello's insight [-@caramello2016bridges; -@caramello2021five] is that formally different theories can share the same classifying topos up to geometric equivalence—a relation termed **Morita equivalence**. When $\mathcal{E}_{\mathbb{T}_1} \simeq \mathcal{E}_{\mathbb{T}_2}$, any property expressible as an invariant of the shared topos transfers automatically from $\mathbb{T}_1$ to $\mathbb{T}_2$ without re-proof.

A Chain of Morita Equivalences Connects the Four Case Theories I

Each Case Framework as a Geometric Theory

We formalize each case-theoretic framework as a geometric theory:

1. **Typological case theory** $\mathbb{T}_{\text{typ}}$: Objects are case roles, morphisms are grammatical relations, axioms specify alignment constraints (e.g., “S = A” in accusative alignment).
2. **Type-logical case theory** $\mathbb{T}_{\text{log}}$: Objects are syntactic types, morphisms are Lambek calculus derivations, axioms specify well-formedness conditions (pregroup contractions).
3. **Distributional case theory** $\mathbb{T}_{\text{dist}}$: Objects are vector spaces, morphisms are linear maps, axioms specify the composition law for distributional meaning (DisCoCat).
4. **Enriched case theory** $\mathbb{T}_{\text{enr}}$: Objects are expressions, morphisms carry $[0, 1]$ -valued weights, axioms specify the identity and composition inequalities.

A Chain of Morita Equivalences Connects the Four Case Theories II

The Bridge Programme: A Chain of Classifying Toposes for Case Theory

The **research programme** we pursue is that these four perspectives admit a topos-theoretic alignment: formally distinct case theories should be related by bridge toposes and, where Morita equivalence can be established, invariants proved in one formulation transfer without re-proof. Equation (1) sketches the *target* picture—a chain of classifying toposes linked by intermediate geometric theories—not a single theorem asserted for all details in this manuscript.

$$\mathcal{E}_{\mathbb{T}_{\text{typ}}} \leftarrow \mathcal{E}_{\mathbb{T}_{\text{bridge}}} \rightarrow \mathcal{E}_{\mathbb{T}_{\text{log}}} \leftarrow \mathcal{E}_{\mathbb{T}_{\text{bridge}}'} \rightarrow \mathcal{E}_{\mathbb{T}_{\text{dist}}} \quad (1)$$

Intermediate toposes would be supplied by geometric theories simultaneously interpretable in both flanking frameworks. *Were* such Morita equivalences established, one would expect:

- **Syntactic theorems port to semantics:** Commutativity results in the type-logical setting would align with

Reconciling Classical DAIF with Intuitionistic Topos Logic via Sheaf Cohomology I

An unresolved tension arises when binding topos-theoretic invariants of case structures with the Distributional Active Inference (DAIF) layer (??). DAIF relies on Quantile Temporal Difference (TD) learning, which assumes Markovian updates over classical probability distributions.

Conversely, the internal logic of a classifying topos is generally intuitionistic—yielding *non-distributive lattices* of truth values where the classical law of excluded middle fails. Forcing classical Markovian density tracking onto non-distributive topological spaces risks mathematical collapse: the case alignments (modeled as geometric properties) may fail to compose covariantly under quantum NLP (lambeq Gen II) decoherence.

Reconciling Classical DAIF with Intuitionistic Topos Logic via Sheaf Cohomology II

To resolve this, we propose that classical DAIF distributions cannot be mapped directly into the topos. Instead, a **sheaf-theoretic bridge** is required: probability densities over case assignments must be treated as *sections of a probability sheaf*. Local discrepancies in case assignment (e.g., during pragmatic garden-path discourse) resolve globally via *sheaf cohomology*. This ensures that while local parses resolve classically via quantile Huber loss, their global composition respects the intuitionistic structure of the geometric case theory.

Phillips's Result: Language-of-Thought Properties as Universal Topos Constructions I

Phillips [-@phillips2024lot] provides a striking application of topos-theoretic methods to cognitive science. He shows that the **Language of Thought** (LoT) hypothesis—the claim that cognition operates over structured, combinatorial representations with language-like properties—can be formalized categorically, and that the resulting structure is *universal* in the topos-theoretic sense.

Specifically, Phillips demonstrates that:

1. LoT properties (discrete constituents, role-filler independence, systematicity) arise as **universal constructions** in a topos—categorical products, fiber bundles, and presheaves.
2. Every topos supports an internal first-order logic, explaining how LoT-like logical capacities can emerge in systems (biological or artificial) whose internal architecture forms a topos.

Phillips's Result: Language-of-Thought Properties as Universal Topos Constructions II

3. The “shape” of cognitive representations is fundamentally **topological**, captured by presheaves and fiber bundles rather than by point-set structures.

Applied to case theory, Phillips's result is significant: because the Language of Thought is topos-universal, and our case categories are definable within any topos (as small categories governed by first-order axioms), every cognitive system with LoT-like architecture has the structural capacity to represent case assignments. This grounds the claim that case structure is a universal feature of higher cognition in the mathematical framework of topos theory rather than in typological observation alone.

Caramello's Syntactic Learning Algorithm: Inducing Case Theories from Annotated Corpora I

Caramello [-@caramello2023syntactic] extends the bridge technique to a learning theory: she shows that classifying toposes can be used to *learn* the theory of a mathematical structure from finite data (a finite set of models). The learning algorithm constructs a classifying topos from the observed data and then extracts the axioms of the underlying theory.

Applied to case systems, this suggests a principled approach to *grammatical induction*: given a corpus annotated with case labels, one could construct the classifying topos of the implicit case theory and read off its axioms—recovering the alignment type, the morphism structure, and the enriched weights from data alone. The procedure operates in four phases:

Caramello's Syntactic Learning Algorithm: Inducing Case Theories from Annotated Corpora II

1. **Extraction:** Parse a Universal Dependencies treebank for a target language, collecting all case-labeled dependency arcs. Each arc (r_1, rel, r_2) instantiates a morphism $r_1 \rightarrow r_2$ in the implicit case category.
2. **Saturation:** Close the extracted morphism set under composition, identity, and the enriched weight constraints of $??$. Compute the empirical hom-values as normalized co-occurrence frequencies.
3. **Classification:** Construct the classifying topos $\mathcal{E}_{\mathbb{T}}$ from the saturated theory—the canonical topos whose models are exactly the case-assignment patterns attested in the corpus. The topos-theoretic invariants (sort count, arity distribution, axiom count) combined with categorical magnitude and magnitude homology ($??$)—computed from the $[0, 1]$ -enriched hom-values of $??$ —provide a multi-dimensional fingerprint of the language's case system.

Caramello's Syntactic Learning Algorithm: Inducing Case Theories from Annotated Corpora III

4. **Identification:** Compare the fingerprint against the Morita equivalence classes of known alignment types. If the classifying topos matches an existing class, the language's alignment type is identified; if not, the procedure has discovered a novel alignment pattern.

This topos-theoretic learning procedure would be provably correct (recovering the true theory in the limit) and maximally general (not presupposing any particular alignment type). The scalar magnitude invariant from ?? enters at step 3 as a summary of the learned category's "effective size," while magnitude homology provides a finer-grained topological signature.

Morita Equivalence Diagrams Are Themselves Free-Ride Inferences I

The bridge technique has a natural diagrammatic interpretation. Morita equivalence between theories is witnessed by *functorial translations*—diagrams in the 2-category of toposes that commute up to natural isomorphism. These diagrams serve the same cognitive function as the commutative diagrams of **??**: they make the transfer of structure visible, allowing a researcher to verify at a glance that a result proved in one framework genuinely applies in another.

Morita Equivalence Diagrams Are Themselves Free-Ride Inferences II

Manders [Manders2008euclidean] observed that even in classical mathematics, diagrams serve not merely as illustrations but as *inferential instruments* whose spatial properties encode proof-relevant information. The topos-theoretic bridge diagrams extend this observation to the meta-theoretical level: the commutative diagram expressing Morita equivalence is itself a “free ride” inference, automatically transferring any topos-invariant property from one theory to another without requiring a case-by-case verification.

Morita Equivalence Diagrams Are Themselves Free-Ride Inferences III

Illustrative transfer (conditional on a bridge). The transitive pregroup derivation $n \cdot (n^r \cdot s \cdot n^l) \cdot n \rightarrow s$ yields the same sentence type whether contractions are grouped left-to-right or right-to-left. That type-logical commutativity is mirrored in DisCoCat by functoriality: the sentence vector for a fixed derivation is well-defined. A full Morita story would package such facts as invariants of a shared classifying topos; here we use the example only to show *what kind* of statement the bridge programme is meant to align across $\mathbb{T}_{\text{log}}$, $\mathbb{T}_{\text{dist}}$, and enriched formulations—not as a claim that every step of (1) is already proved for our case theories.

Python Implementation: Proxy Invariant Checks (**implemented and tested**) I

The topos-theoretic narrative above is paired with a topos Python module that implements **finite geometric theories** extracted from `CaseCategory`, **invariant-profile comparison** (`check_morita_equivalence`), and **guarded bridge transfer** when profiles match—not a full classifying-topos construction inside the runtime.

Python Implementation: Proxy Invariant Checks (implemented and tested) II

Extracting Geometric Theories from CaseCategory Instances

The `build_typological_theory()` function constructs a geometric theory $\mathbb{T}_{\text{typ}}$ from a `CaseCategory` by extracting:

- ▶ **Sorts:** the case roles (objects of the category)
- ▶ **Function symbols:** the morphisms with their source/target pairs
- ▶ **Axioms:** identity morphism existence, composition closure, and alignment constraints

For the standard 8-case category, this yields a theory with 8 sorts and approximately 15 function symbols. The minimal 3-case category produces a theory with 3 sorts and 5 function symbols. Our `build_enriched_theory()` function further annotates the geometric theory with $[0, 1]$ -valued hom weights from the enriched structure of ??.

Python Implementation: Proxy Invariant Checks (implemented and tested) III

ClassifyingTopos Invariants and the Morita Equivalence Check

The `ClassifyingTopos` class computes topological invariants—number of sorts, function arity distribution, and axiom count—that characterize the “logical shape” of a theory. Two theories are **Morita equivalent** when their classifying toposes share the same invariant profile:

```
T_std = build_typological_theory(standard_case_category())
T_min = build_typological_theory(minimal_case_category())
equivalent = check_morita_equivalence(T_std, T_min)
# False: 8-sort and 3-sort theories have different invariants
```

Python Implementation: Proxy Invariant Checks

(implemented and tested) IV

Concrete Morita Equivalence: A Two-Object Illustration

To build intuition, we work out a minimal example. Consider two presentations of the same relational structure — a binary agent-patient dependency — formalized as geometric theories over the two-case system $\{\text{NOM}, \text{ACC}\}$:

Theory $\mathbb{T}_{\text{syn}}$ (syntactic presentation): Sorts $\{n, a\}$; one binary relation symbol $\text{acts_on}(n, a)$; one axiom asserting that every element of sort n participates in at least one acts_on instance.

Theory $\mathbb{T}_{\text{sem}}$ (semantic presentation): Sorts $\{+agent, +patient\}$; one binary relation symbol transfers_force ; the same participation axiom expressed using the semantic sort names.

Both theories present the same classifying topos: the category of sheaves over a two-node directed graph with a single edge. Their invariant profiles match — two sorts, one function symbol of arity $(1, 1)$, one existential axiom — so `check_morita_equivalence` returns `True`: